

ABERDARON, United KingdomWMO: **034050**Lat: **52.7889N**Lon: **4.7414W**Elev: **94**StdP: **100.20**Time Zone: **0.00 (EUW)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-0.2	1.0	-4.7	2.6	1.8	-3.3	2.9	2.6	24.3	8.9	22.3	8.6	6.1	50	0.820

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	3.9	21.4	17.1	19.6	16.2	18.1	15.4	17.8	20.1	16.7	18.5	16.1	17.4	4.1	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.8	12.1	18.7	16.1	11.5	17.3	15.5	11.1	16.7	50.4	20.3	47.3	18.7	45.3	17.5	25.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 21.0	18.3	15.9	-1.6	24.4	1.9	1.9	-2.9	25.8	-4.0	26.9	-5.0	28.0	-6.4	29.4		
(5)			-2.4	19.9	1.7	1.8	-3.7	21.2	-4.7	22.3	-5.7	23.3	-6.9	24.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	10.5	6.3	6.1	6.9	8.6	11.1	13.4	15.0	15.3	14.1	11.9	9.1	7.2
	DBStd	3.94	2.45	2.25	2.18	2.15	2.22	1.97	1.82	1.50	1.59	2.02	2.36	2.60
	HDD10.0	521	114	108	98	51	11	0	0	0	7	42	89	89
	HDD18.3	2883	372	341	355	292	226	149	107	95	126	199	276	344
	CDD10.0	686	0	0	2	9	44	102	154	166	124	66	16	3
	CDD18.3	7	0	0	0	0	0	1	3	2	0	0	0	0
	CDH23.3	13	0	0	0	0	0	3	7	4	0	0	0	0
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0
(14) Wind	WSAvg	7.7	9.7	9.1	7.7	6.6	6.7	6.2	6.0	6.4	7.2	8.7	9.2	9.7
(15) Precipitation	PrecAvg	989	99	71	65	64	61	57	63	78	84	107	122	116
	PrecMax	1276	181	140	131	123	144	170	131	136	138	219	238	315
	PrecMin	769	5	10	9	17	14	15	9	5	13	20	51	24
	PrecStd	132	44	33	30	30	31	39	30	29	33	53	45	58
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.2	11.1	14.1	18.0	20.8	23.0	24.2	23.7	20.3	17.0	14.2	12.5
		MCWB	10.6	9.2	9.8	12.5	15.9	17.2	18.8	19.8	16.6	14.7	13.0	11.6
	2%	DB	10.6	10.1	11.2	14.6	18.0	20.2	21.4	20.5	18.3	15.7	13.5	11.8
		MCWB	10.0	9.1	8.9	11.0	14.4	16.1	17.2	17.2	15.7	14.2	12.6	11.1
	5%	DB	10.0	9.6	10.1	12.4	16.1	18.0	19.2	18.6	17.1	15.1	12.9	11.2
		MCWB	9.3	8.7	8.7	9.9	13.2	14.9	16.1	16.2	15.1	14.0	12.1	10.5
	10%	DB	9.5	9.0	9.4	11.1	14.2	16.4	17.6	17.4	16.3	14.6	12.3	10.7
		MCWB	8.8	8.2	8.3	9.3	11.8	14.1	15.3	15.5	14.6	13.5	11.4	9.9
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.9	10.2	10.5	12.9	16.7	18.4	19.9	20.0	17.6	15.6	13.7	12.0
		MCDB	11.0	10.4	12.3	16.8	19.9	21.7	22.3	23.2	18.8	16.4	13.9	12.2
	2%	WB	10.3	9.5	9.7	11.4	14.9	16.5	17.7	17.7	16.3	14.9	13.0	11.4
		MCDB	10.5	9.8	10.5	13.7	17.5	19.3	20.4	19.6	17.3	15.3	13.3	11.7
	5%	WB	9.6	8.9	9.1	10.5	13.5	15.3	16.6	16.6	15.7	14.3	12.3	10.8
		MCDB	9.9	9.3	9.9	11.9	15.6	17.3	18.5	18.0	16.6	14.8	12.7	11.1
	10%	WB	9.1	8.4	8.6	9.7	12.4	14.4	15.7	16.0	15.1	13.8	11.6	10.1
		MCDB	9.4	8.8	9.3	10.8	13.7	16.0	17.2	16.9	15.9	14.4	12.2	10.6
(35) Mean Daily Temperature Range	5% DB	MDBR	3.0	3.2	3.7	4.3	4.5	4.5	4.2	3.9	3.8	3.3	3.1	3.0
		MCDBR	3.0	3.0	4.1	6.0	7.1	7.4	7.1	5.5	4.6	3.4	3.0	3.2
	5% WB	MCWBR	3.7	3.2	3.3	4.0	4.6	4.9	4.2	3.4	3.3	3.2	3.5	4.0
		MCWBR	3.0	2.9	3.7	5.0	6.3	6.3	5.9	4.6	4.0	3.1	2.9	3.2
(40) Clear-Sky Solar Irradiance	taub		0.329	0.340	0.366	0.381	0.387	0.378	0.382	0.366	0.361	0.348	0.326	0.318
		taud	2.333	2.398	2.360	2.347	2.382	2.432	2.424	2.482	2.490	2.469	2.395	2.307
	Ebn at Noon		646	767	819	852	864	874	864	860	820	749	649	582
		Edh at Noon	62	77	97	111	112	108	107	96	85	71	57	53
(44) All-Sky Solar Radiation	RadAvg		0.68	1.41	2.64	4.18	5.26	5.66	5.12	4.30	3.10	1.72	0.84	0.50
	RadStd		0.07	0.13	0.19	0.28	0.36	0.48	0.46	0.28	0.20	0.15	0.08	0.05

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (44 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page